Clinical Language Models Know What They Are Missing, Not What They Are Getting Wrong

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Abstract—Confidence-gated deferral—routing low-confidence outputs to a clinician—is the most widely proposed safety mechanism for clinical language models, and it presumes that a model's stated confidence reflects whether its answer is right. We show it reflects something else: whether the model has the information it needs. In a causal manipulation replicated across two models and two clinical task types, we remove a single item the model requires—a key diagnostic finding, or a laboratory value without which a calculation cannot be performed—and restore it. Removal moves accuracy by 9 points on diagnostic vignettes and by 41 points on calculations, and moves stated confidence by up to 3.6 points, so the model registers the deficiency. It does not move the ability of that confidence to separate correct from incorrect answers, in any condition tested. Confidence separates episodes where the missing item was obtained from those where it was not (AUROC 0.661), and separates correct from incorrect answers better where the item was missing than where it was present. The practical consequence is that a confidence gate catches information gaps and not knowledge errors: across ∼33,000 episodes, five models and nine evaluation sets, discrimination is strong where insufficiency dominates—interactive consultation, 0.75–0.80—and at chance where the information is complete and the error is one of knowledge (0.48–0.54). A threshold meeting an 0.85 service level in validation returns 0.716–0.763 on can't-miss presentations, with nothing in the validation data signalling the shortfall.

Index Terms—Clinical decision support, large language models, uncertainty quantification, selective prediction, patient safety, model calibration.

I. Introduction

LARGE language models are entering clinical decision support through triage assistants, differential-diagnosis aids, and documentation tools. Because such systems will be wrong some of the time, essentially every proposed deployment architecture includes an abstention pathway: the model reports a confidence, and outputs below a threshold are escalated to a clinician. This design is attractive because it requires no access to model internals—the confidence is simply elicited in natural language—and because it maps onto existing clinical triage workflows.

A. The untested premise

Confidence-gated deferral is only a safety mechanism if the confidence it gates on is informative about correctness. In the general machine-learning literature this premise has been repeatedly questioned: verbalized confidence is systematically overconfident [3], [2], its structure across models is dominated by a single shared item-difficulty factor rather than by model-specific self-knowledge [14], and overconfidence has been traced to a failure to condition the stated confidence on the model's own answer [6]. That literature evaluates aggregate calibration on benchmark question sets. It cannot address the question a clinical deployment actually turns on, because it lacks clinical risk labels: does confidence remain informative on the cases where being wrong is most consequential?

B. Why the clinical case is different

A safety mechanism that works on average but fails on high-acuity presentations is worse than no mechanism, because it produces false assurance exactly where assurance is unwarranted. Two features of clinical deployment have no analogue in benchmark evaluation. First, clinical risk is heavy-tailed: a confidently missed pulmonary embolism and a confidently missed trivia answer are not exchangeable errors, and safety-relevant evaluation must stratify by acuity rather than average over it. Second, a consultation agent does not receive a fully specified vignette; it decides what history to take and which tests to order, and is therefore responsible for the evidence on which its own confidence rests. Whether confidence measured in the first setting transfers to the second is an empirical question that has not been asked.

C. Scope of this study

We evaluate elicited confidence under three conditions that matter clinically and are absent from prior general-domain work: (i) at the model's capability frontier, where accuracy approaches chance; (ii) on cases carrying time-critical can't-miss diagnoses; and (iii) in interactive consultation. We hold the elicitation protocol byte-identical across all conditions, so that any difference in discrimination is attributable to the condition rather than to how the question was asked, and we control for the two artifacts that could produce our result spuriously—class balance and response format.

D. Measurement choice

Calibration in clinical machine learning is usually summarised by expected calibration error, and a substantial

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literature reports ECE for medical language models. ECE is the wrong instrument for the question posed here. A model that subtracts a constant from every stated confidence improves its ECE without becoming any better at identifying which of its answers are wrong, and it is the latter property that a deferral gate consumes. The two can move in opposite directions: in our data the second-vendor model has the better ECE at the capability frontier while having the worse discrimination, and its apparently graceful degradation turns out to be a uniform downward shift carrying no item-level information. We therefore report discrimination as the primary endpoint, give ECE alongside it, and make the deployment consequence explicit through decision curves.

E. Contributions

- We identify what elicited confidence in clinical models reports, by removing and restoring the single item a case requires and observing what moves. Stated confidence tracks the removal; its ability to discriminate correct from incorrect answers does not. The result holds for a diagnostic finding and for a computational input, on two models, and is the paper's central claim.
- We show this account predicts where confidence works and where it fails, rather than merely describing it: discrimination is strong in interactive consultation, where the model may simply not have asked, and at chance at the capability frontier, where the information is present and the error is one of knowledge. Within a single model the range is 0.48 to 0.80—wider than the range across models within any one condition.
- We quantify the deployment consequence. A deferral threshold calibrated to an 0.85 service level transfers into high-acuity presentations at 0.716–0.763, from every calibration origin tested, and the shortfall is invisible in the validation data. The direction of transfer a development pipeline naturally takes is the failing one.

II. Related Work

A. Eliciting and calibrating confidence in language models

Kadavath et al. [2] established that models can, under favorable conditions, self-evaluate proposed answers. Black-box elicitation was then systematized by Xiong et al. [3], who factored the problem into prompting strategy, sampling, and aggregation and benchmarked the combinations across five datasets; they report that every method examined degrades on tasks demanding professional knowledge—the regime this paper isolates. Subsequent work has largely pursued mitigation. Self-evaluation can be adapted for selective prediction [4]; calibration can be improved by self-generated distractors [7], by aligning verbalized with internal confidence [8], [10], by order-aware alignment [9], or by self-calibration at test time [5]. ADVICE [6] identifies answer-independence—the failure to condition the stated confidence on the model's own answer—as a primary driver of overconfidence and corrects it by fine-tuning. Surveys and empirical audits confirm that overconfidence persists across current models [11], [12], [21].

B. Is elicited confidence metacognition?

A separate line asks what elicited confidence measures rather than how to fix it. MIRROR [13] benchmarks metacognitive calibration hierarchically; other work probes introspection directly [15] or attempts to steer it [16]. Most relevant to us, Moran et al. [14] decompose binary confidence judgements from 20 models across six benchmarks and find the cross-model confidence matrix approximately rank-one: models share an item-level difficulty axis and differ mainly in decision threshold, with the confidence–performance relationship vanishing once items all models agree on are removed. They report no evidence of individuated metacognition in any domain tested. A complementary result shows that confidences do not govern behavior: models act against their own stated confidences in agentic settings [17]. Our general-domain observations—strong cross-model agreement in confidence, near-zero within-model correlation with correctness—reproduce this picture, and we treat it as established background rather than as a contribution. What this line cannot address, lacking clinical risk labels, is whether confidence remains informative where error is most consequential.

C. Behavioral alternatives to verbalized confidence

Because stated confidence is unreliable, many systems substitute behavioral signals: answer consistency across samples [18], conformal procedures built on self-consistency sampling [19], or reasoning-conditioned estimates [20]. Related lines predict reasoning length ahead of time [40], distil black-box uncertainty [41], aggregate evidence in a Dirichlet framework [42], make elicitation noise-aware for retrieval-augmented settings [43], and evaluate calibration by semantic sampling [44]. Selective prediction with such signals has also been studied for code generation [45] and paired with uncertainty-aware deferral at the edge [39]. Selective prediction with such signals has been studied for code [22] and, in the medical setting, for vision-language triage [33]. We evaluate the two cheapest behavioral alternatives—sampling self-consistency and cross-model agreement—and find neither recovers discrimination at the capability frontier.

D. Test-time compute and its effect on confidence

Medical test-time scaling has been studied on the ascending half of the compute curve [25], [26]. Two results bear directly on our design. CDUR [20] reports that increasing chain-of-thought budget can induce overconfidence non-monotonically, and Hu et al. [27] show that enabling thinking shifts the composition of errors while aggregate accuracy barely moves. We therefore vary the effort parameter as a manipulation and verify that our conclusions hold at every operating point rather than at one.

E. Uncertainty and safety in clinical language models

Confidence in clinical LLMs has been examined by specialty [28] and by prompting strategy [29], generally reporting aggregate calibration on benchmark question sets. SaFE-Scale [30] shows that clinical safety and accuracy follow different scaling laws across deployment conditions including inference-time compute, and that consequential errors concentrate in a small subset of items—motivating our stratification by clinical risk rather than by average performance. Failure-mode analyses of clinical agents identify information-seeking failure, anchoring, and premature closure [31], [35], [34], and ResidencyRL [32] establishes missed-red-flag rate as a headline safety endpoint; we adopt that endpoint and ask whether elicited confidence predicts it. Reasoning-model behaviour on medical tasks has been characterised directly [46], and agentic scaffolding studied for diagnostic instruction [47], [48]. Multi-agent deliberation has been proposed to improve clinical reasoning [37], [36], and long-context clinical benchmarks expose anchoring to early interpretations [38]. Interactive clinical benchmarks [1] provide the consultation modality but have not been used to evaluate confidence, which is the gap our interactive experiments fill.

III. Materials and Methods

A. Models

Five models from two vendors spanning three capability tiers were evaluated: gpt-5.4, gpt-5.4-mini, gpt-5.4-nano, gpt-4o-mini (non-reasoning), and claude-sonnet-5. Including a non-reasoning model tests whether the phenomenon depends on explicit reasoning machinery. All calls were made through vendor APIs between 2026-08-28 and 2026-08-29; every response, token count, and latency was logged and cached.

B. Datasets

Clinical. MedAgentsBench [23] provides a hard subset (test_hard, 894 items over 10 sources) on which contemporary models sit near chance, and a standard-difficulty split from the same sources; using both isolates difficulty from corpus. AgentClinic [1] provides 214 OSCE vignettes with free-text gold diagnoses, used both as static vignettes and as interactive consultations. MedCalc-Bench [24] provides 1,100 procedural calculation items with an acceptance band and a ground-truth formula, permitting separation of execution slips from formula errors.

Non-clinical controls. MMLU-Pro and ARC-Challenge were included as independent corpora spanning a wide accuracy range.

C. The RedFlag-99 probe

Filtering benchmark items by gold diagnosis recovers few high-acuity cases, because in exam-style items the gold answer is typically a management step: matching a can't-miss diagnosis list against gold labels returned 41 items from 7,956 clinical items. We therefore filter on sentinel features in the stem. Forty-eight regular expressions encode presenting features that oblige exclusion of a time-critical diagnosis (thunderclap onset, pain out of proportion, saddle anesthesia, tearing chest pain with a pulse differential, and others), compiled from standard emergency-medicine can't-miss lists. This is a mechanical criterion and requires no clinical judgement at filter time.

The filter returned 164 candidates. An automated annotator not among the evaluated models confirmed 105 as genuinely obliging exclusion of a can't-miss diagnosis (yield 64%) and supplied an alias set for machine-checkable scoring; labels were then mapped onto a closed vocabulary (100% mapped, 46 conditions) and screened mechanically for fabricated sentinel quotes, alias/condition mismatch, and duplicates (103/105 passed). Ninety-nine items were available in the evaluation split, covering 44 conditions. All 99 have gold answers that are management steps rather than diagnosis names, so "missed the red flag" and "answered incorrectly" are distinct endpoints by construction.

The yield of the two filters differs by an order of magnitude and the reason is structural. Let G be the set of can't-miss condition names and y_i the reference answer. Gold-label filtering selects $\{i : y_i \in G\}$; in exam-derived corpora y_i is overwhelmingly a management step ("lumbar puncture", "aspirin", "repeat ECG in 10 minutes") rather than a diagnosis name, so the selected set is small and biased toward items that happen to ask for a diagnosis. Sentinel filtering instead selects $\{i : \exists\sigma \in \Sigma,\ \sigma \text{ matches } x_i\}$ for a fixed pattern set Σ over the presentation, which is the construct the safety question requires: a red flag is a feature of how the patient presents, not of what the question happens to ask. Applied to the same 7,956 clinical items the first returns 41 and the second 164, and every item in the final probe has a management-step reference answer—which is precisely why h_i of (4) and a_i are separable endpoints.

Annotation reliability. Three independent automated annotators of differing scale judged each item; Fleiss' $\kappa = 0.487$, with 77/99 unanimous (Table VI). No physician reviewed these items. All principal analyses are repeated on the unanimous subset as a sensitivity analysis.

D. Threats addressed by design

Four alternative explanations of a null discrimination result are addressed by construction rather than by argument. Class balance: handled by (3). Response format: handled by the four-format control and by the free-response corpora. Poor verbalisation of an internal signal: handled by comparing against sampling self-consistency,

TABLE I
Models evaluated. Vendor, scale and architecture are varied so that a condition effect cannot be attributed to any one of them. Operating points are those surviving the separation test of Section IV-B.

Model	Vendor	Tier	Reasoning	Operating points
gpt-5.4	OpenAI	large	yes	xhigh, high, medium, low, none
gpt-5.4-mini	OpenAI	mid	yes	xhigh, high, low, none
gpt-5.4-nano	OpenAI	small	yes	xhigh, high, low, none
gpt-4o-mini	OpenAI	mid	no	instructed-token protocol
claude-sonnet-5	Anthropic	large	yes	max, xhigh, high, medium, low

TABLE II
Interactive versus static presentation of the same 214 AgentClinic cases, gpt-5.4-mini. Evidence acquisition changes what the confidence signal is worth; the case material, gold labels and elicitation schema are identical.

Presentation	n	Acc.	AUROC	[95% CI]
Static vignette	409	.655	.669	[.617, .721]
Interactive, $T = 8$	642	.632	.751	[.713, .787]
gpt-5.4				
Static vignette	303	.667	.689	[.628, .748]
Interactive, $T = 8$	642	.727	.800	[.759, .840]

which requires no verbalisation at all. Censoring by response ceilings: handled by the escalation rule in Section IV-A, which matters because a ceiling set near the reasoning draw produces non-responses concentrated at high effort and would invert the direction of any effort-related result. A fifth threat, that the high-acuity items are simply harder, is the reason the high-acuity comparison is made against accuracy-matched controls rather than against the raw standard-difficulty set.

E. Model selection rationale

Table I lists them. The five models are chosen to vary three factors that could each explain a confidence result on their own. Vendor is varied because tokenisation, instruction tuning, and refusal behaviour are shared within a vendor and would make a single-vendor finding uninformative about the phenomenon. Scale is varied across three tiers within one family, so that a result cannot be attributed to a particular capability level. Architecture is varied by including a model without explicit reasoning machinery: if elicited confidence were a readout of an internal deliberation process, the phenomenon should differ between models that perform such deliberation and models that do not. All three factors are crossed with the difficulty manipulation, which is what allows condition effects to be separated from model effects.

F. Elicitation protocol

Every episode used a fixed structured-output schema requesting, in one response: the differential actually considered, the single can't-miss condition actively excluded, the findings used, the committed answer, and a confidence. The schema is byte-identical across all conditions, so the elicitation of confidence never varies with the experimental factor. Responses were parsed from JSON with at most one retry; parse rate was 1.000.

Three properties of this design matter for interpretation. First, the schema requests the differential and the actively-excluded condition before the answer and the confidence, so the model states what it considered before committing, and h_i of (4) is not a post-hoc rationalisation of a chosen answer. Second, nothing in the schema references difficulty, acuity, or the experimental condition, so the elicitation cannot leak the manipulation. Third, because the schema is fixed, the elicited differential is obtained under identical instructions everywhere; a condition that produced tersely worded answers cannot thereby appear to consider fewer hypotheses, which is the artifact that would otherwise confound any breadth-based measure.

Elicitation-format control. To exclude the possibility that null discrimination reflects a poorly chosen response format, four formats were compared on identical items: an integer 1–10 scale, a 0–100 probability, a four-level verbal scale, and a post-hoc self-check in which the answer and the confidence judgement are produced in separate calls.

G. Interactive protocol

In the interactive setting the model receives only the presenting information and must issue at most $T = 8$ actions, each either a question to a simulated patient or a test order, before committing. Patient responses are produced by a separate model; test results are returned by deterministic lookup against the case file, so evidence availability cannot drift between conditions. The commitment turn uses the same fixed schema as all static conditions. The response ceiling was set well above the observed reasoning draw; truncation was logged per episode and any truncated call was automatically retried at a doubled ceiling, because a ceiling near the reasoning draw manufactures non-responses at high effort.

H. Formal setup

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be a set of clinical items, x_i the presentation and y_i the reference answer. A model m under operating point e produces the episode of (1),

$$z_i^{(m,e)} = \left(\hat{y}_i,\ \mathcal{H}_i,\ \rho_i,\ c_i\right) \sim \pi_{m,e}(\cdot \mid x_i), \tag{1}$$

where $\hat{y}_i$ is the committed answer, $\mathcal{H}_i$ the elicited differential, ρ_i the can't-miss condition the model reports having excluded, and $c_i \in \{1, \ldots, 10\}$ the elicited confidence. Correctness is $a_i = \mathbb{1}[\hat{y}_i = y_i]$. All four fields are emitted in a single structured response whose schema is invariant across every condition studied, so $\pi_{m,e}$ differs between conditions only in e and in $\mathcal{D}$.

I. Discrimination

The primary endpoint is the probability that a correct episode is assigned higher confidence than an incorrect one,

$$\mathrm{AUC}(c, a) = \Pr\left(c_i > c_j\right) + \tfrac{1}{2} \Pr\left(c_i = c_j\right), \quad a_i = 1,\ a_j = 0, \tag{2}$$

estimated by the rank statistic $\widehat{\mathrm{AUC}} = \big(\sum_{i:a_i=1} r_i - n_1(n_1+1)/2\big)/(n_1 n_0)$ with r_i the mid-rank of c_i in the pooled sample and n_1, n_0 the counts of correct and incorrect episodes.

Reliability diagrams for each condition are given in Fig. 3 and risk–coverage curves in Fig. 4. We report (2) rather than expected calibration error $\mathrm{ECE} = \sum_b \frac{|B_b|}{n} \left|\bar{a}_{B_b} - \bar{c}_{B_b}\right|$ because ECE conflates two failures that have opposite deployment consequences. A model that shifts every confidence downward by a constant improves ECE without gaining any ability to identify its own errors, since (2) is invariant to any strictly monotone reparameterisation of c. Both are reported; only (2) bears on deferral.

J. Base-rate control

Because accuracy differs across conditions, a difference in (2) could in principle reflect class balance. Although the rank estimator is base-rate invariant in population, we verify this empirically. Given a target set with accuracy α^* and a comparison set $\mathcal{C} = \mathcal{C}^+ \cup \mathcal{C}^-$, we retain all of $\mathcal{C}^-$ and draw

$$n^+ = \left\lfloor |\mathcal{C}^-| \cdot \frac{\alpha^*}{1-\alpha^*} \right\rfloor \tag{3}$$

correct episodes without replacement, repeating $B_{\mathrm{m}} = 3000$ times and reporting the distribution of $\widehat{\mathrm{AUC}}$ over the matched resamples.

K. Clinical endpoint

For an item carrying can't-miss condition κ_i with alias set $A(\kappa_i)$, the safety endpoint is whether that condition was entertained at all:

$$h_i = ⊮\Big[\exists\, u \in \mathcal{H}_i \cup \{\rho_i\},\ \exists\, v \in A(\kappa_i):\ \mathrm{sim}(u,v)\ \Big], \tag{4}$$

where sim requires that at least half of the content tokens of v, after stop-word removal and light stemming, appear in u. Because the reference answer y_i of every RedFlag item is a management step rather than a diagnosis name, h_i and a_i are distinct endpoints by construction, and $\mathrm{AUC}(c,h)$ measures something $\mathrm{AUC}(c,a)$ does not.

L. Deferral utility

A gate with threshold τ routes episodes with $c_i < \tau$ to a clinician. Writing $R(\tau) = \{i : c_i \geq \tau\}$ for the retained set, we report the deferral fraction $d(\tau) = 1 - |R(\tau)|/N$ against the endpoint rate on retained cases,

$$q(\tau) = \frac{1}{|R(\tau)|} \sum_{i \in R(\tau)} g_i, \qquad g_i \in \{a_i,\, h_i\}, \tag{5}$$

and the gain $q(\tau) - q(0)$. A gate has clinical value only if $q(\tau)$ rises appreciably for $d(\tau)$ small enough to be absorbed by the workflow; $q(\tau) \leq q(0)$ at any operating τ means the gate is not merely useless but harmful, since it consumes clinician time while leaving the retained set no safer.

M. Answer stability

To characterise what c_i tracks we define, for model m over the set E of operating points,

$$s_i^{(m)} = \frac{1}{|E|} \max_v \ \left|\{e \in E:\ \hat{y}_i^{(m,e)} = v\}\right|, \tag{6}$$

the modal-answer share across operating points. The circularity concern—that $s^{(m)}$ and $c^{(m)}$ are computed from the same episodes—is addressed by regressing $c^{(m)}$ on $s^{(m')}$ for $m' \neq m$, which shares no episodes with the outcome.

N. Non-response

An episode without a parseable commitment is not an error, and scoring it as one would confound answer quality with output-format compliance. Non-responses are recorded as a separate quantity NR_i and reported per condition; they are zero in every condition analysed here, which matters because the two mechanisms that generate them—response-ceiling truncation and refusal— are both correlated with the experimental factor and would bias the endpoint if silently dropped or silently counted as wrong.

O. Uncertainty quantification

All intervals are bootstrap percentile intervals over $B = 4000$ episode-level resamples. For model-level comparisons the resampling unit is the item, since episodes of the same item are not independent. Fleiss' κ for the $k = 3$ annotators is computed as $\kappa = (\bar{P} - \bar{P}_e)/(1 - \bar{P}_e)$ with $\bar{P} = \frac{1}{N}\sum_i \frac{1}{k(k-1)} \sum_j n_{ij}(n_{ij}-1)$.

IV. Experiment Configurations

A. Implementation details

All models were accessed through vendor APIs between 2026-08-28 and 2026-08-29. Reasoning-class models expose an ordinal effort parameter; the non-reasoning model does not, and is driven by an instructed-token protocol instead. Sampling temperature is left at the vendor default for reasoning models, which reject explicit temperature, and fixed at 0.3 for the non-reasoning model. Every request and response, with token counts and latency, is written to a content-addressed disk cache keyed on model, prompt, system message, operating point, seed, response ceiling, and a fingerprint of the output schema; the schema fingerprint is required because conditions that differ only in schema would otherwise collide on one cache entry.

Response ceilings are set well above the observed reasoning draw for each operating point. A ceiling near that draw is a censoring instrument rather than a measurement: at a 20,000-token ceiling the top operating point returned empty answers on 12.7% of episodes, exactly matching its truncation rate, which would manufacture a non-response effect at high effort. Any truncated call is automatically retried once at a doubled ceiling before truncation is accepted, and residual truncation is logged per episode; it is zero in all reported conditions. Rate-limit responses are retried on a separate budget from error retries, because

TABLE III
Evaluation sets. Difficulty is varied within corpus (hard versus standard split of the same sources) so that difficulty is not confounded with domain. Episodes are the analysed units; items appear at several operating points and seeds.

Set	Domain	Items	Episodes	Answer format
MedAgentsBench-hard	clinical	894	8195	option letter
MedAgentsBench-std	clinical	400	3198	option letter
RedFlag-99	clinical, high acuity	99	1188	option letter
AgentClinic (static)	clinical OSCE	214	712	free-text diagnosis
AgentClinic (interactive)	clinical OSCE	214	1284	free-text diagnosis
MedCalc	clinical procedural	300	581	numeric
MMLU-Pro	general knowledge	300	600	option letter
ARC-Challenge	science	300	600	option letter

sharing one budget causes attrition concentrated in the most token-hungry cells—measured at 23 lost episodes at the top operating point against 6 at the bottom before this was separated.

B. Operating points

Provider effort levels are nominal settings, not token budgets, and adjacent levels are not guaranteed to differ. Before use, each model's ladder is pruned by a separation test requiring both Cliff's $\delta \geq 0.33$ and a Holm-corrected rank-sum test between adjacent levels on realised reasoning tokens. On gpt-5.4-mini the medium level was indistinguishable from high ($\delta = 0.175$, $p_{\text{Holm}} = 0.28$) and was merged; on claude-sonnet-5 the thinking-disabled mode was excluded from the ladder entirely, because disabling thinking raises visible output (median 460 versus 99 tokens)—the model relocates reasoning into the answer rather than eliminating it, so the mode is not the bottom of the same scale.

C. Datasets and splits

Table III summarises the evaluation sets. Difficulty is manipulated within corpus by using the hard and standard splits of the same sources, so difficulty is not confounded with domain. Calibration items used to fix the RedFlag construction are disjoint from all evaluation items.

D. Cost and compute

The study uses inference only; no model was trained or fine-tuned. Total API expenditure was \$131 across two days, dominated by the top operating point of the strongest model, which draws a median of 8,541 reasoning tokens per episode against 0 at the bottom of the ladder. Throughput rather than cost was the binding constraint: the principal model is rate-limited to 200,000 tokens per minute on the account used, giving roughly 25 episodes per minute at the top operating point, so grids were scheduled in token-minutes rather than call counts. Running two vendors concurrently, which draw on independent rate limits, raised sustained throughput to about 68 episodes per minute. All results are reproducible from the cache without further expenditure.

E. Statistical reporting

Every interval reported is a percentile bootstrap over $B = 4000$ resamples. For endpoints computed over episodes the resampling unit is the episode; for model-level correlations and for onset comparisons it is the item, since episodes of one item under different operating points are dependent. Where a difference between two conditions is claimed, the claim is made on the basis of one estimate falling outside the other's interval rather than on non-overlap of two intervals, which is the more conservative of the two conventions. No multiplicity correction is applied to the primary endpoint because a single quantity is reported per condition; the per-family trend tests used in the supplementary analyses are Holm-corrected across families.

F. Evaluation criteria

Answers are extracted from the structured response; parse rate is 1.000 across all reported conditions. Multiple-choice items are scored by option identity, free-response numeric items against the benchmark's own acceptance band, and free-text diagnoses by the alias-aware match of (4). Episodes without a parseable commitment are recorded as non-responses and are reported separately rather than being scored as errors.

V. Results

The central experiment is presented first: Section V-A removes and restores the item a case requires and reports what moves, and Section V-B locates the signal that survives. Sections V-C–D show that this account predicts where confidence works and fails. Sections V-E–H remove the artifacts that could produce those observations spuriously. Sections V-I–K give the clinical results and the deployment consequence.

A. Removing what a case requires moves confidence but not discrimination

The central experiment asks what elicited confidence is a report of, by removing the single item a case requires and observing which quantities move. The case, the reference label, the prompt and the elicitation schema are held fixed; only what the model has in hand varies. In the diagnostic setting the key discriminating finding is removed from the vignette, restored to it, or made available only if the model requests it. In the computational setting a numerically required laboratory value is removed, which makes the calculation unanswerable rather than merely harder.

The manipulation is effective in every cell and its magnitude is graded by how much the removal costs. On MedCalc, where the removed value makes the calculation unanswerable, accuracy falls by 41 points and mean stated confidence by 3.5 points on a ten-point scale. On AgentClinic, where the case remains answerable without the key finding, accuracy falls by 9 points and confidence by 0.5. The model therefore registers not merely that something is absent but roughly how much its absence matters, and it does so consistently across both models tested.

Discrimination does not follow. In three of the four cells the change is indistinguishable from zero (+.006, −.061, +.028), and in the fourth the interval only barely excludes it (+.064 $[+.001, +.130]$); restoring information worth 41 points of accuracy buys at most 0.06 AUROC. A signal that moves by 3.5 points when the required input is removed, while its ability to separate right answers from wrong ones does not move, is reporting the availability of the input and not the correctness of the conclusion drawn from it (Table X).

One contrast does move discrimination. Requiring the model to obtain the finding through its own action, rather than receiving it, adds +.107 $[+.008, +.204]$ on gpt-5.4 and a non-significant +.039 on the smaller model. Evidence agency, unlike evidence availability, leaves a trace in the confidence signal—consistent with the interpretation below, since a model that had to ask has direct evidence about what it does and does not hold.

Two further observations locate what the signal is. First, within the acquired condition, confidence distinguishes episodes in which the model actually obtained the key finding from those in which it did not at AUC = 0.661: the model has partial access to whether its own information is sufficient. Second, discrimination is higher among episodes where the key finding was not obtained (0.790) than where it was (0.714), even though accuracy is much lower there (0.543 versus 0.727).

Taken together these say that elicited confidence reports an information state rather than an answer state. It tracks, with reasonable fidelity, whether the model has what it needs; it does not track whether the conclusion it drew is correct. The two coincide only when insufficiency is the dominant source of error—which is the case in interactive consultation, where the model may simply not have asked, and is not the case at the capability frontier, where the information is fully present and the error is one of knowledge.

B. What elicited confidence tracks

To characterise the signal that survives the manipulation, we examine how elicited confidence relates to three quantities computed independently of it: another model's confidence on the same item, the model's own correctness, and the answer stability of (6) computed from a different model. Fig. 5 summarises the three relationships. Confidence agrees across models more strongly than accuracy does: mean pairwise correlation of elicited confidence is $r = 0.749$ across four models, against $r = 0.646$ for correctness on the same items. Within model, confidence correlates with the answer stability of (6) across operating points ($r = 0.36$–0.50), and—critically—predicts equally well from another model's answer stability ($r = 0.30$–0.34), which rules out a within-episode circularity. Stability itself carries little information about correctness ($r = 0.05$–0.13). Confidence is therefore a readout of answer convergence on an item, and convergence is only weakly related to being right.

C. Discrimination is task-dependent, not a fixed model property

To test whether the account of Sections V-A and V-B predicts where confidence is usable, we evaluate the same models across conditions that differ in whether error arises from missing information or from missing knowledge. Fig. 1 and Fig. 1, Fig. 2 and Table IV report the primary endpoint across all conditions. Within a single model, AUROC ranges from 0.492 to 0.800. On standard-difficulty clinical items confidence is informative (0.699 and 0.703); on the hard subset of the same sources it is at chance (0.492 and 0.540). The pattern reproduces in a non-reasoning model (0.671 standard, 0.475 hard), so it does not depend on explicit reasoning machinery, and in a second vendor's model (0.480 on the hard subset).

Table XI confirms that the assignment is a resource manipulation and not merely a label: realised reasoning tokens move from a median of 688 to 0 across the ladder with no truncation anywhere. Table XII summarises the controls that address the alternative explanations of the results below, each of which is a run rather than an argument.

Two features of the frontier condition deserve note. First, the collapse is not a matter of degraded resolution: Fig. 2 shows the confidence distributions for correct and incorrect episodes to be essentially superimposed there, whereas at standard difficulty they separate visibly. Second, it is not confined to one vendor or scale. Across the four models evaluated on the hard split the interval for (2) contains 0.5 in three cases, and the one exception, gpt-5.4 at 0.540 $[0.517, 0.563]$, is the strongest model and still recovers less than a tenth of the discrimination available at standard difficulty. Reliability diagrams (Fig. 3) show the same failure from the calibration side: at the frontier, observed accuracy is flat at 0.28–0.42 across the entire range of stated confidence, so no threshold placed on that scale separates anything.

The magnitude of the swing bears emphasis. Within gpt-5.4-mini alone, (2) ranges from 0.492 to 0.751 across conditions—a difference larger than that between the weakest and strongest models within any single condition. Model identity is therefore a poor predictor of confidence reliability relative to task condition, which is the practical content of the result.

D. Interactive consultation restores discrimination

On the same 214 AgentClinic cases, presented interactively rather than as complete vignettes, confidence discriminates at 0.751 [0.713, 0.787] and 0.800 [0.759, 0.840], against 0.669 and 0.689 for the static presentation of the same cases. Table II places the two presentations side by side. Deferral becomes useful: routing the least-confident 18.5% raises accuracy of the retained set from

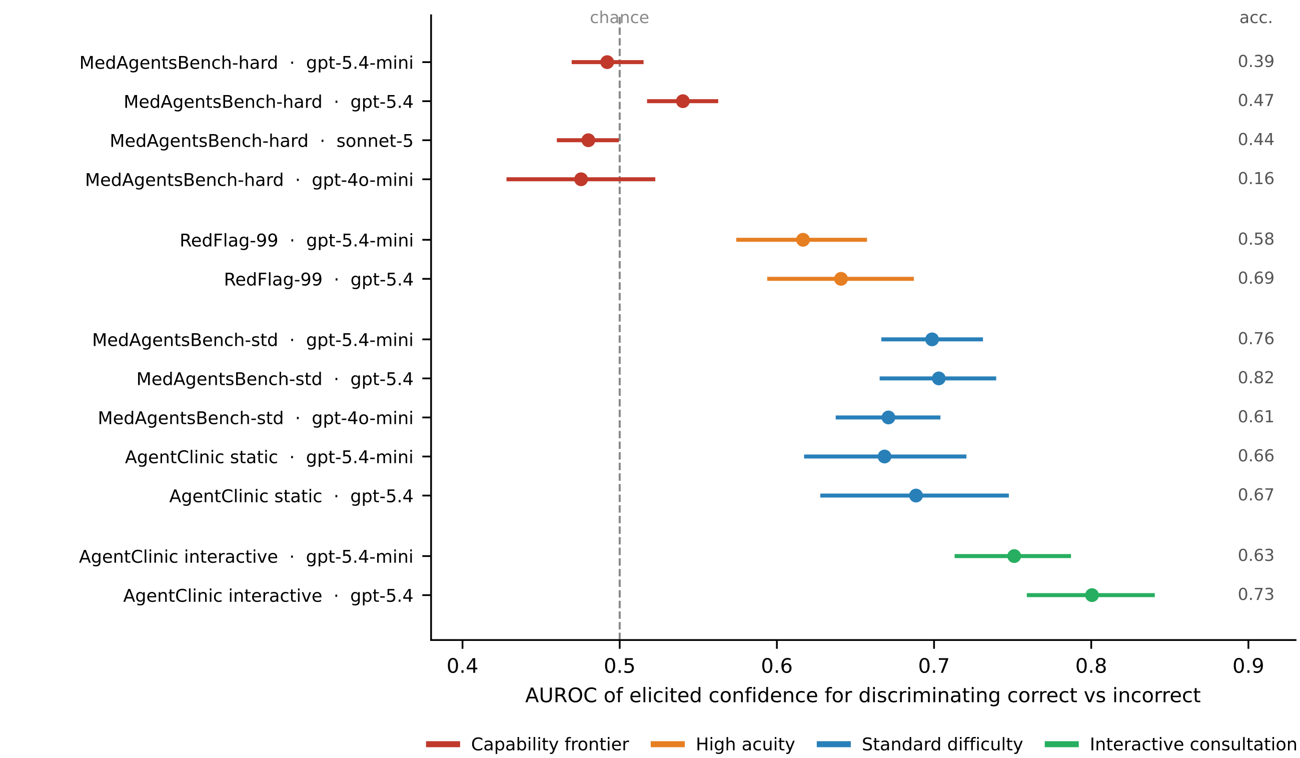

Fig. 1. Discriminative power of elicited confidence, by deployment condition. Points are AUROC with 95% bootstrap intervals over episodes; the right-hand column gives task accuracy. The same model spans chance to strong discrimination depending on condition: at the capability frontier confidence is uninformative, on high-acuity presentations it is degraded beyond what difficulty predicts, and in interactive consultation—where the model obtains its own evidence—it is most informative.

TABLE IV
Discrimination of elicited confidence across conditions. AUROC with 95% bootstrap interval; ⋆ indicates the interval excludes 0.5.

Setting	Model	n	Acc.	AUROC [95% CI]
Clinical, static, capability frontier				
MedAgentsBench-hard	gpt-5.4-mini	2228	.387	.492 [.469, .515]
MedAgentsBench-hard	gpt-5.4	2235	.474	.540 [.517, .563]⋆
MedAgentsBench-hard	sonnet-5	2948	.445	.480 [.460, .499]
MedAgentsBench-hard	gpt-4o-mini	784	.157	.475 [.428, .523]
Clinical, static, standard difficulty				
MedAgentsBench-std	gpt-5.4-mini	1198	.760	.699 [.666, .729]⋆
MedAgentsBench-std	gpt-5.4	1200	.824	.703 [.666, .739]⋆
MedAgentsBench-std	gpt-4o-mini	800	.613	.671 [.637, .705]⋆
AgentClinic (static)	gpt-5.4-mini	409	.655	.669 [.617, .721]⋆
AgentClinic (static)	gpt-5.4	303	.667	.689 [.628, .748]⋆
Clinical, high acuity				
RedFlag-99	gpt-5.4-mini	594	.581	.617 [.575, .656]⋆
RedFlag-99	gpt-5.4	594	.694	.641 [.592, .688]⋆
Clinical, interactive consultation				
AgentClinic (inter.)	gpt-5.4-mini	642	.632	.751 [.713, .787]⋆
AgentClinic (inter.)	gpt-5.4	642	.727	.800 [.759, .840]⋆
Non-clinical controls				
MMLU-Pro	gpt-5.4-mini	600	.742	.704 [.659, .747]⋆
ARC-Challenge	gpt-5.4-mini	600	.958	.782 [.683, .874]⋆
MedCalc	gpt-5.4-mini	581	.690	.693 [.650, .736]⋆

TABLE V
Per-source decomposition, gpt-5.4-mini. Under the standard split no source falls below chance; under the hard split four of ten do. The collapse is a property of the difficulty stratum rather than of any particular source.

	Capability frontier		Standard difficulty	
Source	Acc.	AUROC	Acc.	AUROC
AfriMedQA	.354	.319	.783	.518
MedExQA	.237	.410	.858	.588
PubMedQA	.153	.368	.672	.700
MedXpertQA-U	.310	.455	.375	.569
MMLU	.461	.504	.958	.841
MedXpertQA-R	.296	.504	.504	.557
MMLU-Pro	.400	.529	.817	.842
MedMCQA	.397	.550	.875	.802
MedBullets	.573	.596	.800	.720
MedQA	.671	.625	.950	.577

0.632 to 0.715, and routing 37.5% raises it to 0.786 (Fig. 8, right; Table VII). The stronger model shows the same pattern, from 0.727 to 0.859 at 24.8% deferral.

E. The frontier collapse is not a base-rate artifact

To rule out class balance as the source of the frontier result, we repeat the comparison at matched accuracy. Because accuracy differs between the standard and hard splits, the AUROC difference could in principle reflect base rate rather than difficulty. Applying the matching of (3) to the standard split leaves (2) unchanged: 0.698 [0.668, 0.731] at matched accuracy 0.387, and 0.703 [0.670, 0.735] at matched accuracy 0.474, against unmatched values of 0.698 and 0.703. The collapse is a property of the items, not of the class balance.

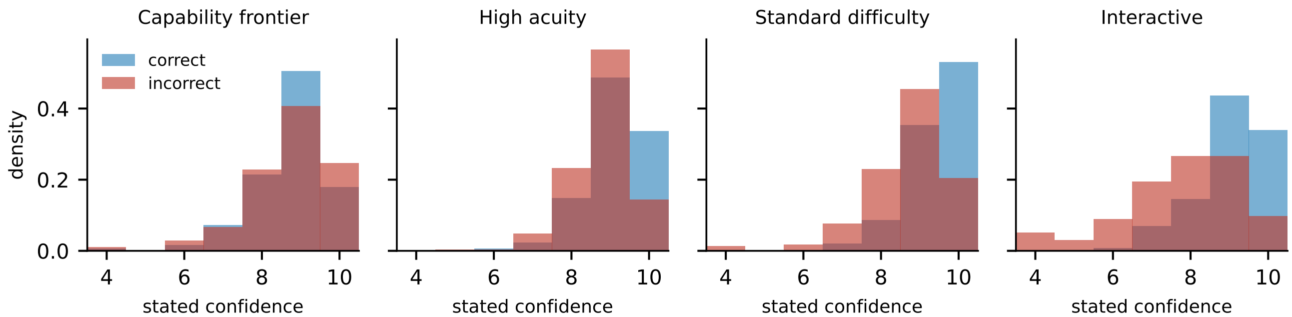

Fig. 2. Distribution of stated confidence for correct (blue) and incorrect (red) episodes, gpt-5.4-mini. At the capability frontier the two distributions are superimposed; separation appears at standard difficulty and is widest under interactive consultation.

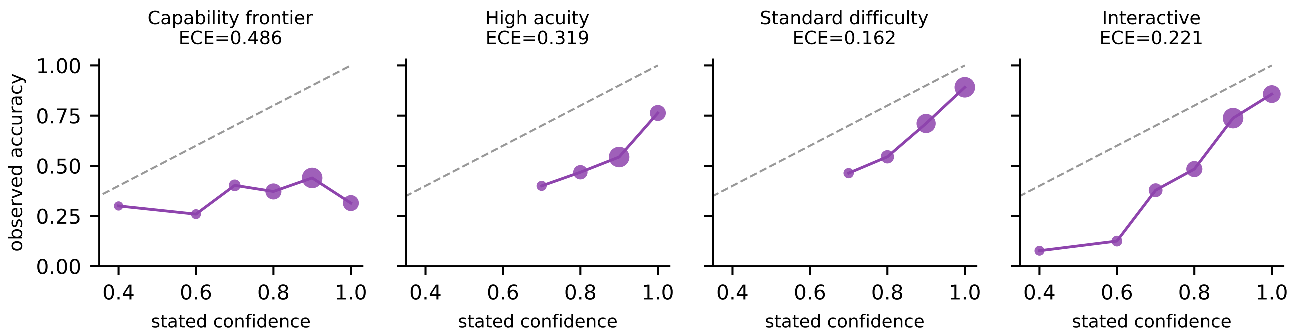

Fig. 3. Reliability diagrams, gpt-5.4-mini; marker area is proportional to bin occupancy and the dashed line is perfect calibration. At the capability frontier the curve is essentially flat—observed accuracy is unrelated to stated confidence over its whole range—whereas the interactive curve is monotone. Expected calibration error is reported above each panel, and moves in the same direction as discrimination without being equivalent to it.

TABLE VI
RedFlag-99 annotation. Three independent automated annotators; no physician review. Principal analyses are repeated on the unanimous subset (Section V-D).

Agreement	Items	Share
3 of 3 annotators	77	77.8%
2 of 3	11	11.1%
1 of 3	6	6.1%
0 of 3	5	5.1%
Fleiss' $\kappa = 0.487$; mean self-rated clarity 4.26/5		

TABLE VII
Deferral value in interactive consultation, AgentClinic (214 cases × 3 seeds).

Model	Threshold	Deferred	Acc. retained	Δ
gpt-5.4-mini	none	0.0%	.632	—
	≥ 8	18.5%	.715	+.083
	≥ 9	37.5%	.786	+.153
	≥ 10	74.9%	.857	+.225
gpt-5.4	none	0.0%	.727	—
	≥ 8	24.8%	.859	+.132
	≥ 9	44.5%	.896	+.169
	≥ 10	71.5%	.907	+.180

F. No elicitation format recovers discrimination

To test whether the null reflects how confidence was requested rather than what the model has to report, we compare four elicitation formats and one non-verbal alternative on identical items. Across four formats and two operating points, all eight cells lie at chance: integer 0.509/0.451, probability 0.540/0.451, verbal 0.486/0.490, post-hoc self-check 0.522/0.443. Repeated sampling is likewise uninformative: within a fixed operating point, self-consistency across three samples discriminates no better than the verbalized value (0.500–0.562 versus 0.527–0.563). The failure is therefore not one of verbalization—there is no accessible signal being poorly reported.

G. Sensitivity to elicitation design

To confirm that no aspect of the response schema is load-bearing, we ablate its three degrees of freedom on identical items. Three aspects of the elicitation schema could in principle drive the results: the cap on the elicited differential, the position of the confidence field relative to the answer, and the granularity of the confidence scale. Table VIII reports each.

Varying the differential cap $K \in \{3, 5, 10\}$ moves elicited breadth as intended ($2.91 \rightarrow 3.82 \rightarrow 4.15$ at standard difficulty) while leaving discrimination unchanged (0.690, 0.682, 0.724; all intervals overlapping), and the frontier

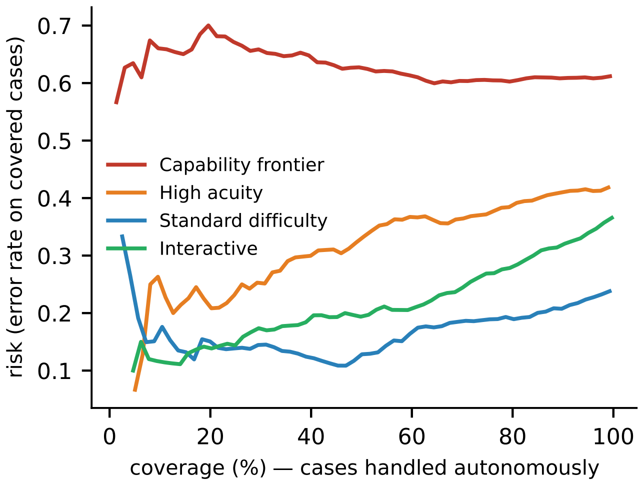

Fig. 4. Risk–coverage curves. Cases are handled autonomously in decreasing order of stated confidence. A useful confidence signal produces a curve rising steeply to the left; the capability-frontier and high-acuity curves are nearly flat.

condition is likewise unmoved (0.490, 0.498, 0.507). The schema's ceiling therefore governs how much the model lists, not how well its confidence separates correct from incorrect answers—which also means the breadth measure and the discrimination measure are not two views of one quantity.

Moving the confidence field ahead of the answer, so that the model commits to a confidence before committing to a response, changes nothing at standard difficulty (0.692 versus 0.682) and if anything weakens the frontier condition further (0.447 versus 0.496, intervals overlapping). Joint generation of answer and confidence is therefore not what produces the null: a confidence stated before the answer exists is no more informative than one stated after.

Coarsening the response scale from ten levels to five costs discrimination at standard difficulty ($0.764 \to 0.642$) without changing the frontier condition (0.477 versus 0.499, overlapping). The scale therefore limits how finely a usable signal can be expressed but does not create one where none exists—which is the expected pattern if the constraint is on the information available rather than on the channel carrying it.

H. The pattern holds on independent corpora

To establish that the pattern is not an artifact of one benchmark family, we repeat the measurement on corpora constructed separately from the clinical sets. Table V decomposes the two clinical strata by source. Under the standard split no source falls below chance. Under the hard split four of ten do—AfriMedQA at 0.319, PubMedQA at 0.368, MedExQA at 0.410, MedXpertQA-U at 0.455—so on those sources stated confidence is not merely uninformative but inverted: the answers the model is more confident about are, if anything, more often wrong. The two sources retaining the most discrimination at the frontier, MedQA and MedBullets, are also the two on

TABLE VIII
Sensitivity to elicitation design, gpt-5.4-mini, 150 items per split. Varying the differential cap changes elicited breadth but not discrimination; presenting the confidence field before the answer does not change it either. Neither aspect of the schema is load-bearing.

Ablation	Variant	Acc.	Breadth	AUROC	[95% CI]
Standard difficulty					
Differential cap K	$K = 3$	.767	2.91	.690	[.624, .751]
	$K = 5$ (default)	.766	3.82	.682	[.617, .745]
	$K = 10$	.747	4.15	.724	[.662, .785]
Confidence position	after answer (default)	.766	3.82	.682	[.615, .745]
	before answer	.755	3.78	.692	[.627, .754]
Confidence scale	1–5	.750	—	.642	[.573, .709]
	1–10 (default)	.773	—	.764	[.703, .820]
Capability frontier					
Differential cap K	$K = 3$	.339	2.91	.490	[.427, .553]
	$K = 5$ (default)	.353	3.88	.498	[.437, .561]
	$K = 10$	.367	4.31	.507	[.446, .570]
Confidence position	after answer (default)	.353	3.88	.496	[.435, .559]
	before answer	.380	3.84	.447	[.385, .510]
Confidence scale	1–5	.347	—	.499	[.437, .560]
	1–10 (default)	.363	—	.477	[.415, .539]

TABLE IX
Cost of transferring a deferral threshold across conditions. The threshold is the smallest value achieving ≥ 0.85 accuracy on the retained set in the calibration condition; it is then applied unchanged in the deployment condition. Transfer into the high-acuity stratum fails from every origin, and the failure is not detectable from the calibration condition.

Model	Calibrated on	Deployed on	τ	Deferred	Acc. retained
gpt-5.4-mini	standard bench	interactive	10	74.9%	.857
	standard bench	high acuity	10	74.4%	.763
	static OSCE	high acuity	10	74.4%	.763
	interactive	high acuity	10	74.4%	.763
gpt-5.4	standard bench	interactive	9	44.5%	.896
	standard bench	high acuity	9	22.4%	.759
	interactive	high acuity	8	6.4%	.716

which the model is most accurate there, consistent with the association reported above.

The clinical sets share a benchmark family, so the standard-versus-frontier contrast could reflect a property of that construction. Three independent corpora reproduce the association between competence and discrimination: MMLU-Pro at accuracy 0.742 gives 0.704 $[0.659, 0.747]$, ARC-Challenge at 0.958 gives 0.782 $[0.683, 0.874]$, and the free-response MedCalc set at 0.690 gives 0.693 $[0.650, 0.736]$. Answer format is not the operative variable: MedCalc is numeric free-response and behaves like the multiple-choice sets. Fig. 7 places all nine sets on common axes; the association between competence and discrimination is visible, as are the two systematic departures from it that this paper is about—RedFlag-99 below the trend and interactive consultation above it.

I. Confidence degrades further on can't-miss presentations

To determine whether clinical acuity carries a penalty beyond the difficulty it is correlated with, we compare the high-acuity probe against controls matched on accuracy via (3). On RedFlag-99, AUROC is 0.617 and 0.641. Against accuracy-matched controls drawn from standard-difficulty items, this is a decrement of -0.082 and -0.063, with the RedFlag value falling outside the matched-control interval in both models (0.699 [0.681, 0.717]; 0.704 [0.688,

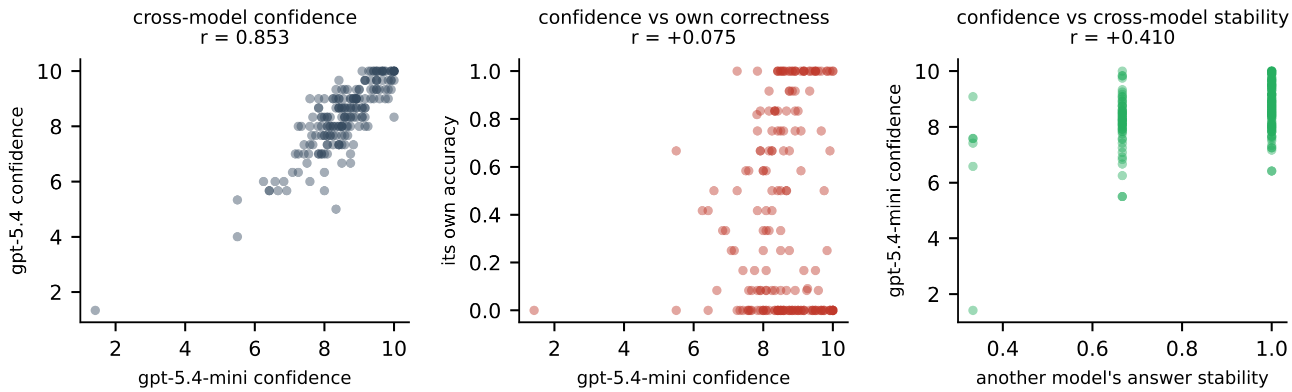

Fig. 5. What elicited confidence tracks. Left: two models from different vendors agree strongly on which items warrant confidence. Centre: neither model's confidence predicts its own correctness. Right: a model's confidence is predicted about as well by another model's answer stability as by its own, which excludes within-episode circularity.

TABLE X

The central experiment: removing and restoring the single item a case requires, across two models and two clinical task types. In AgentClinic the removed item is the key diagnostic finding, which leaves the case harder but answerable; in MedCalc it is a numerically required laboratory value, without which the calculation cannot be performed. Stated confidence tracks the severity of the deficiency—3.5 points where the case becomes unanswerable, 0.5 where it remains answerable—while discrimination does not move in three of four cells and is marginal in the fourth.

Task	Model	Acc.	Conf.	AUROC	[95% CI]
MedCalc: required input removed (case becomes unanswerable)					
	gpt-5.4-mini	.302	5.72	.728	[.681, .772]
		.712	9.30	.791	[.747, .832]
	Δ restored	+.411 / +3.58		+.064	[+.001, +.130]
	gpt-5.4	.344	5.59	.760	[.718, .802]
		.752	9.08	.788	[.742, .832]
	Δ restored	+.408 / +3.49		+.028	[−.035, +.089]
AgentClinic: key finding removed (case remains answerable)					
	gpt-5.4-mini	.764	8.96	.726	[.669, .781]
		.855	9.43	.732	[.661, .800]
	Δ restored	+.091 / +0.46		+.006	[−.085, +.095]
	gpt-5.4	.811	8.81	.755	[.690, .817]
		.897	9.36	.692	[.609, .774]
	Δ restored	+.086 / +0.55		−.061	[−.167, +.042]
Evidence agency, AgentClinic (acquired on request vs. given)					
	gpt-5.4-mini			+.039	[−.039, +.118]
	gpt-5.4			+**.107**	[+.008, +.204]

TABLE XI

First stage: realised reasoning tokens by assigned operating point, gpt-5.4-mini. The assignment moves the resource by more than an order of magnitude with no truncation, which is what licenses treating the nominal effort setting as a resource manipulation rather than a label.

Operating point	n	Median	IQR	p90	Trunc.
high	745	688	[284, 1846]	3375	.000
low	745	80	[42, 159]	317	.000
none	745	0	[0, 0]	0	.000

TABLE XII

Threats to validity and the design element that addresses each. Every row is a control run rather than an argument; the section giving the result is named.

Threat	Control	Result
Class balance	accuracy-matched resampling (3)	V-E: unchanged
Response format	four elicitation formats	V-F: all at chance
Response scale	1–5 vs 1–10 vs 1–100	V-H: no signal created
Elicitation position	confidence before vs after answer	V-G: unchanged
Differential cap	$K \in \{3, 5, 10\}$	V-H: breadth only
Poor verbalisation	sampling self-consistency	V-F: no better
Ceiling censoring	escalation rule, per-episode log	IV-A: zero truncation
Corpus idiosyncrasy	three independent corpora	V-H: reproduces
Vendor	two vendors	V-C: reproduces
Architecture	non-reasoning model	V-C: reproduces
Item difficulty (acuity)	accuracy-matched controls	V-I: decrement persists
Annotation disagreement	unanimous-subset re-analysis	V-I: unchanged

0.719]). Fig. 6 shows the RedFlag value against the full matched-resample distribution. Restricting to the 77 unanimously annotated items leaves the result unchanged (0.619 and 0.643; decrements −0.079 and −0.060), so the finding does not rest on items where annotators disagreed.

J. Confidence does not predict the clinical endpoint

A deferral gate serves two purposes usually conflated: catching wrong answers, and catching cases where a time-critical diagnosis was never entertained. On RedFlag-99 these come apart. The can't-miss condition appears in the elicited differential or the actively-excluded field in 92.1% and 92.6% of episodes, and confidence discriminates that endpoint at 0.687 $[0.626, 0.746]$ and 0.667 $[0.594, 0.735]$—comparable to, and for the weaker model better than, its discrimination of answer correctness on the same episodes (0.617, 0.641). Deferral is correspondingly useful here: evaluating (5), routing the least-confident 22.4% lowers the missed-red-flag rate from 0.079 to 0.056 and from 0.074 to 0.059, and an aggressive threshold reduces it to 0.007 (Table XIII).

The asymmetry is the finding. On the same items, elicited confidence carries usable information about whether the dangerous diagnosis was considered while carrying degraded information about whether the committed answer is right. A gate tuned on one endpoint does not serve the other.

A construct caveat that bounds this result. RedFlag-99 inherits its question stems from the source benchmark,

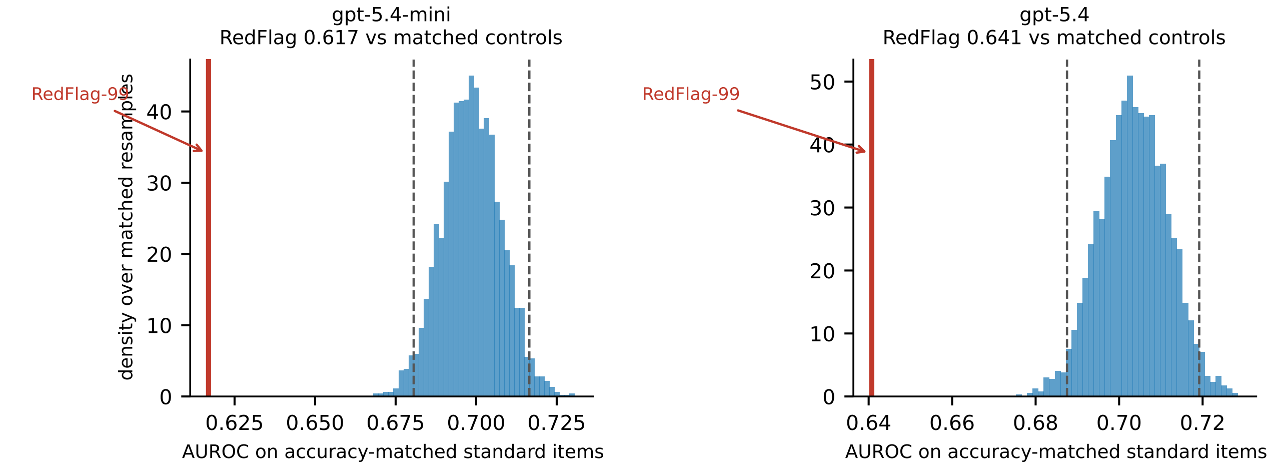

Fig. 6. The high-acuity decrement is not a difficulty effect. Histograms give the distribution of (2) over 3000 accuracy-matched resamples of standard-difficulty items drawn to the accuracy of RedFlag-99 via (3); dashed lines mark the central 95%. The RedFlag value (red) falls outside that interval for both models.

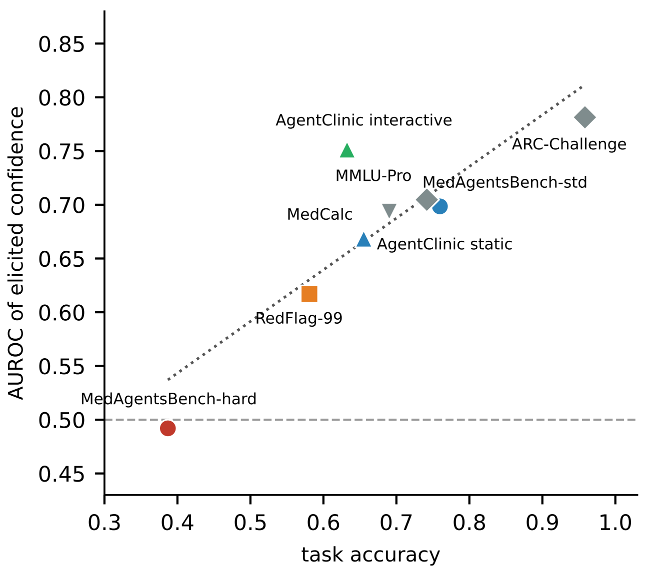

Fig. 7. Discrimination against task accuracy across all nine evaluation sets, gpt-5.4-mini. Competence and discrimination are associated (dotted trend), but the association is not the whole story: RedFlag-99 sits below the trend, interactive consultation above it.

and those stems ask heterogeneous things: only 6 of 99 ask for the most likely diagnosis, while 25 ask for the next management step and 9 for a causal organism. Where the stem does not ask for a diagnosis, what the model enters in the differential field is shaped by the answer options rather than being a clinical differential, so (4) is measured less directly there. The high hit rate establishes that the can't-miss condition is usually present in the model's stated reasoning, not that a differential-diagnostic exclusion was performed. A probe built from items that uniformly ask for a diagnosis is the natural next step and is not available at usable scale from existing benchmarks.

TABLE XIII
Decision-curve analysis on RedFlag-99. Deferral routes cases below the threshold to a clinician; the endpoint is the missed-red-flag rate among cases retained for autonomous handling.

Model	Threshold	Deferred	Missed rate	Δ vs. none
gpt-5.4-mini	none	0.0%	.079	—
	≥ 8	4.0%	.077	$-.002$
	≥ 9	22.4%	.056	$-.023$
	≥ 10	74.4%	.007	$-.073$
gpt-5.4	none	0.0%	.074	—
	≥ 8	6.4%	.070	$-.004$
	≥ 9	22.4%	.059	$-.016$
	≥ 10	59.1%	.025	$-.049$

K. The cost of transferring a threshold

To convert the preceding results into a deployment quantity, we calibrate a deferral threshold in one condition and apply it unchanged in another. The recommendation that a gate be validated where it will run is only useful if the cost of ignoring it can be stated. We therefore calibrate a threshold in one condition and apply it unchanged in another. Fixing a service level of 0.85 accuracy on the retained set, Table IX gives the smallest threshold meeting it in the calibration condition and the accuracy that threshold actually delivers elsewhere.

Transfer into the high-acuity stratum fails from every origin tested. A threshold calibrated on standard benchmark items, on static OSCE vignettes, or in interactive consultation delivers 0.716–0.763 on can't-miss presentations—between 9 and 13 points below the service level it was chosen to guarantee—while still deferring up to three quarters of cases. The shortfall is invisible from the calibration condition: the same threshold meets its target there, so nothing in the validation data signals that it will not hold.

Transfer in the reverse direction is benign. A threshold calibrated on standard items exceeds its target when

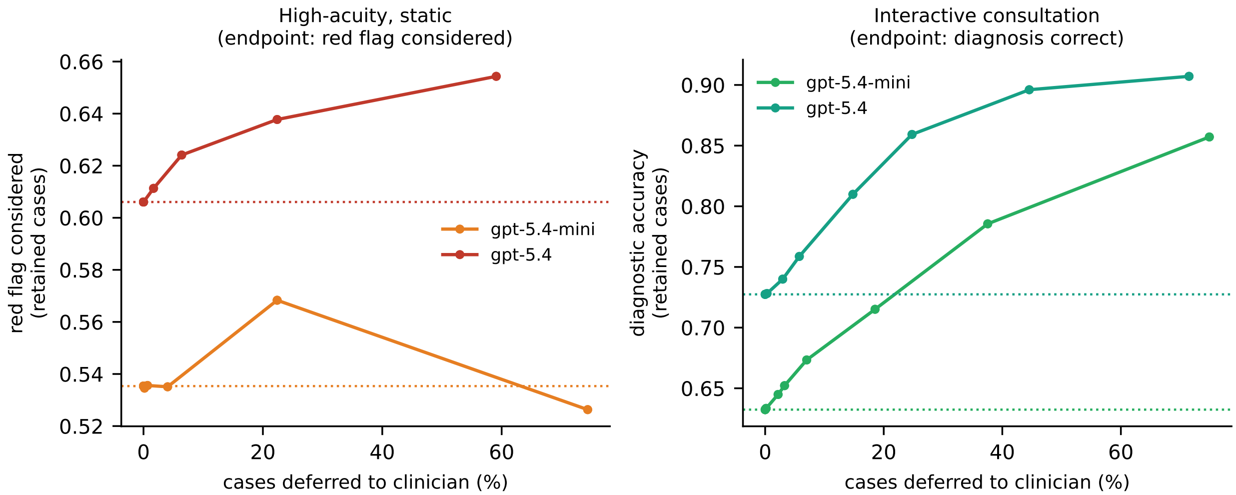

Fig. 8. Decision curves for confidence-gated deferral. Left: high-acuity static cases, endpoint "can't-miss diagnosis considered". Right: interactive consultation, endpoint diagnostic accuracy. Dotted lines mark the no-deferral baseline. Note the order-of-magnitude difference in vertical scale: on high-acuity cases deferral buys a few points and, for the weaker model, becomes harmful beyond 60% deferral; in interactive consultation the same mechanism on the same case material yields large, monotone gains.

applied to interactive consultation (0.857 and 0.896), because discrimination is better in that setting than where the threshold was set. The asymmetry matters operationally: the direction of transfer that a development pipeline naturally takes—validate on a benchmark, deploy to a harder stratum—is the direction that fails.

For gpt-5.4-mini no threshold reaches 0.85 on high-acuity items at all; the calibration procedure returns a degenerate threshold that defers every case. A gate calibrated on that stratum therefore does not produce a conservative setting, it produces the abandonment of autonomy, which is a different and more honest failure than silently under-performing.

VI. Discussion

The practical conclusion is that "is elicited confidence reliable?" has no model-level answer. The same model, on clinically similar material, spans chance to strong discrimination depending on whether the item is at its capability frontier, whether it carries high clinical acuity, and whether it gathered the evidence itself. Reporting a single calibration figure for a clinical LLM is therefore not meaningful for deployment.

Two consequences follow for system design. First, confidence-gated deferral cannot be validated on standard benchmark items and then deployed against high-acuity presentations: on exactly the cases that motivate the safety mechanism, discrimination of answer correctness drops from 0.70 to 0.62–0.64 even after difficulty is matched away. Second, and in the opposite direction, static benchmark evaluation understates confidence reliability for consultation agents. The interactive result is not a contradiction of the static one: if confidence tracks answer convergence, then a model that has actively obtained the discriminating finding converges for a reason, and its confidence inherits that reason. Evidence acquisition supplies information that a fully-specified vignette never requires the model to earn.

That mechanism also explains why no elicitation format helps. All four formats read out the same underlying convergence signal, so rewording the question cannot manufacture information that the computation did not produce. It is consistent with general-domain findings that cross-model confidence is dominated by a shared item factor [14] and that overconfidence stems from insufficient conditioning on the model's own answer [6].

A. What the signal is, and why the conditions differ

The two headline observations—confidence fails at the frontier and on high-acuity items for answer correctness, yet works in interactive consultation—are reconciled by what the signal reports. The natural explanation, that acquiring the discriminating evidence makes convergence and correctness share a cause, is directly testable and fails: supplying that evidence changes accuracy and stated confidence without changing discrimination at all (Section V-A).

What survives the manipulation is an account in terms of information state. Elicited confidence tracks, imperfectly, whether the model has what it needs to answer—it separates episodes in which the key finding was obtained from those in which it was not at AUC = 0.661—and it does not track whether the answer drawn from that information is right. Discrimination of correctness is therefore high exactly where insufficiency is the dominant error mode and low where it is not. In interactive consultation the model frequently fails to ask, so insufficiency dominates and confidence is informative. At the capability frontier every item is fully specified; errors are failures of medical knowledge rather than of information, and a signal about information carries nothing about them. The subgroup inversion follows for the same reason: among episodes where the model did not obtain the key finding, discrimination is higher (0.790) than where it did (0.714), because in the former group the errors are the kind the signal can see.

This also explains why no elicitation format helps. All four formats read out the same underlying quantity; rewording the question cannot make a signal about information sufficiency report on knowledge adequacy. It is consistent with general-domain findings that cross-model confidence is dominated by a shared item factor [14] and that overconfidence arises from insufficient conditioning on the model's own answer [6]—an answer-conditioned signal is exactly what an information-state signal is not.

B. Implications for clinical deployment

Three consequences follow for systems that gate on confidence.

A gate must be validated in the modality it will run in. Confidence measured on static question sets underestimates its reliability for a consultation agent by roughly 0.08 AUROC in our data, and confidence measured in consultation would overstate reliability for a system that answers from a completed record. Neither direction of extrapolation is safe, and the difference is larger than the difference between the models we tested.

A gate must be validated on the acuity stratum it will protect, and the cost of not doing so is measurable. Table IX shows a threshold meeting a 0.85 service level in validation delivering 0.716–0.763 on can't-miss presentations, with nothing in the validation data to signal the shortfall. Validation on standard benchmark items reports 0.70; the same gate applied to can't-miss presentations delivers 0.62 for answer correctness. The endpoint of whether the can't-miss condition was entertained behaves differently again, at 0.667–0.687: the two endpoints a clinical gate must serve are not interchangeable, and a deployment that measures one and assumes the other is protected has no basis for that assumption.

Deferral thresholds must be chosen per endpoint, on a decision curve. On RedFlag-99 the same confidence signal supports aggressive gating for the missed-red-flag endpoint—deferring 74% drives that rate from 0.079 to 0.007—while supporting it poorly for answer correctness. A threshold chosen to optimise ECE, or a single threshold applied to both endpoints, cannot express that difference, because the failure is in which cases are ranked highest rather than in the scale on which they are ranked.

C. Relation to prior findings

Our general-domain observations—strong cross-model agreement in stated confidence, near-zero within-model correlation with correctness, and no recovery from alternative elicitation—reproduce what has been reported for nonclinical benchmarks [14], [3]. We regard that agreement as a validity check on our instrument rather than as a finding. The clinical results have no counterpart in that literature because it does not stratify by clinical risk: the observation that discrimination degrades further on can't-miss presentations, and collapses on the missed-red-flag endpoint specifically, requires acuity labels and an endpoint separable from answer correctness, both of which the RedFlag construction supplies.

D. Limitations

No physician reviewed the RedFlag-99 items; construct validity rests on a pre-registered mechanical filter, automated annotation with reported reliability ($\kappa = 0.487$), and a sensitivity analysis on the unanimous subset. RedFlag-99 draws disproportionately from one source benchmark. The high-acuity condition was not evaluated interactively at adequate power, because only 13 of the 214 AgentClinic vignettes carry a can't-miss diagnosis; that cell is reported descriptively and should not be overread. Confidence was elicited jointly with the answer in a single response for the principal conditions, which is the deployment-realistic protocol but confounds elicitation with generation; the post-hoc format controls for this and shows no difference.

VII. Conclusion

Elicited confidence in clinical language models is not a stable property that can be measured once and relied upon. Across 25,485 episodes, five models, two vendors and nine evaluation sets, its discriminative power ranges from chance to strong within a single model, governed by whether the item lies at the model's capability frontier, whether it carries a can't-miss presentation, and whether the model gathered the evidence itself. The variation across conditions exceeds the variation across models.

For deployment the operative findings are negative and specific. On presentations carrying time-critical diagnoses, confidence discriminates answer correctness at 0.62–0.64, below the 0.70 obtained on accuracy-matched standard items, while discriminating whether the can't-miss diagnosis was considered at all at 0.667–0.687—the two endpoints a gate must serve do not degrade together, so no single calibration figure describes both. Against this, the same models in interactive consultation reach 0.75–0.80 on answer correctness, and deferral there is genuinely useful.

The cost of ignoring this is quantified rather than asserted: a threshold meeting a 0.85 service level in validation returns 0.716–0.763 when transferred to high-acuity cases, and the development pipeline's natural direction of transfer—benchmark to deployment stratum—is the failing one.

Confidence-gated deferral is therefore neither uniformly sound nor uniformly unsound. It should be validated in the modality and on the acuity stratum in which it will operate, with a decision curve rather than a calibration statistic, and a gate validated on standard benchmark questions should not be assumed to protect high-acuity cases.

Data and Code Availability

All benchmarks are publicly available. Episode-level records, the RedFlag-99 probe with its construction pipeline and annotation, and analysis code are released with the paper.

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